JSG Part I (A)

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September 13, 2025

Tenets and Contracts

Standing Hypotheses (H1–H6).

(H1) Decision space Γ is smooth/connected; symmetry by Lie group when needed.

(H2) G is Whitney-stratified; dynamics specified per stratum and prolonged (Filippov) at frontiers.

(H3) Classical identities hold stratumwise; frontier crossings induce a defect budget per crossing.

(H4) Evaluation functionals are Borel and L^2 under invariant/windowed laws.

(H5) EF neighborhood at independence with Fisher $F \succ 0$ and Hessian-Lipschitz L_ψ.

(H6) Controllers are Dirac-projected and band-limited PR on $[\omega_L, \omega_H]$ with margin ϵ.

Symbol Registry (standing constants and moduli)

All constants are understood as ***uniform on compact sublevel sets*** *of the Lyapunov functions in use.* For quick cross-references elsewhere, use `\constref{...}`. *[Symbol Registry table omitted in this build due to a tabularx parsing error; will be restored after CI fix.]*

Acceptance checks for this part.

- Frontier events: logs include N_{cross} and $t/\tau_{\min}$ bound.
- Defect budgets: per-crossing contribution tracked using $\kappa_{\max}, \theta_{\min}, r_{\mathrm{tube}}$.
- PR certificates: band-limited positive-real margin reported ($\geq \epsilon$).
- DRO dual: windowed $\rho = c\,\hat{L}\,\tau$ with $\eta^\star$ recorded.

1 State Spaces and Hybrid Exterior Algebra

This section fixes the geometric setting and proves the *Hybrid Calculus Axioms*. New here: (i) a *No-Zeno* lemma under strict transversality with an explicit dwell time $\tau_{\min}$ (Lemma 2); (ii) a *geometry-scaled* Jacobi-defect bound with curvature/incidence dependence (Lemma 4); (iii) a DEC consistency result with a frontier term controlled by No-Zeno (Proposition 2).

1.1 Whitney stratifications and tangent structures

Definition 1 (Whitney stratified pseudomanifold)**.** Let G be a second-countable, Hausdorff space with a finite disjoint decomposition $G = \bigsqcup_{i\in I} M_i$ into connected, smooth manifolds (strata), satisfying: (i) the frontier condition; (ii) Whitney (a) and (b) regularity for incident pairs (M_i, M_j) with $M_i \prec M_j$. Each M_i carries a smooth Riemannian metric g_i; the collection is a *stratified Riemannian structure.*

When needed we work in an ambient smooth manifold $\widetilde{G}$ and view strata as embedded submanifolds with induced metric and second fundamental form. We write $TG = \bigsqcup_i TM_i$ and denote the tangent *cone* at $x \in \overline{M_i}$ by $C_x M_i$.

Definition 2 (Frontier-transversal vector fields)**.** A family $V = \{V_i\}_{i\in I}$ with $V_i \in \mathfrak{X}(M_i)$ is *frontier-transversal* if for every incident pair $M_i \prec M_j$ and $x \in M_i \cap \overline{M_j}$ there exist extensions $\widetilde{V}_j$ to a tubular neighborhood of M_i in $\widetilde{G}$ such that $\lim_{y\to x,\, y\in M_j} \mathrm{proj}_{T_x M_i} \widetilde{V}_j(y)$ exists and $\lim_{y\to x,\, y\in M_j} \mathrm{proj}_{N_x M_i} \widetilde{V}_j(y) = 0$.

1.2 Filippov selection and No-Zeno

Trajectories may cross strata. We model dynamics by a differential inclusion $\dot{x} \in F(x)$ on $X = \Gamma \times G$ defined stratumwise and prolonged to frontiers by Filippov regularization. Write $x = (\gamma, g)$.

Definition 3 (Filippov regularization on stratified spaces)**.** Let $F_i : \Gamma \times M_i \rightrightarrows T(\Gamma \times M_i)$ be bounded, measurable with closed, convex values and locally one-sided Lipschitz in x. The Filippov regularization $\mathcal{F}$ associates to each $x \in \Gamma \times G$ the convex closure of essential values of F_j in neighborhoods approaching x along strata with x in their closure. A Carathéodory solution is an absolutely continuous curve $x(\cdot)$ with $\dot{x}(t) \in \mathcal{F}(x(t))$ a.e.

Assumption 1 (Non-explosive frontier dynamics)**.** (i) F_i has linear growth and one-sided Lipschitz modulus L_F on each stratum; (ii) there exists a Lyapunov function whose sublevel sets intersect only finitely many strata.

Remark 1 (Normals on Whitney frontiers and failure set)*.* Choose $n(x)$ as the unit normal field induced by a C^1 tubular neighborhood of the embedded frontier representative in the ambient $\widetilde{G}$. Strict transversality may fail on a set of codimension ≥ 1; the set of *crossing times* that hit this failure set has one-dimensional Lebesgue measure 0.

Lemma 1 (Transversality under C^1 perturbations)**.** *There exists a residual (dense G_δ) set of C^1 perturbations of the stratumwise fields for which $\inf_{x\in\Sigma} \inf_{v\in\mathcal{F}(x)} |\langle v, n(x)\rangle| \geq \alpha > 0$. Equivalently, for a generic perturbation the evaluation map $(x, v) \mapsto \langle v, n(x)\rangle$ has 0 as a regular value on Σ, so tangencies occur only on a set of codimension ≥ 1 and do not affect crossing-time measures.* Uses: *Sard-type argument on $\langle v, n\rangle$; C^1 tubular $n(\cdot)$ from Remark 1.*

Lemma 2 (No-Zeno under strict transversality)**.** [Rep/Alg/Stoch] *Let Σ be a smooth frontier with unit normal n. Suppose one-sided Lipschitz per stratum and strict transversality:*

$$\inf_{x\in\Sigma} \inf_{v\in\mathcal{F}(x)} |\langle v, n(x)\rangle| \;\geq\; \alpha > 0.$$

Here $n(x)$ is as in Remark 1; by Lemma 1 the hypothesis is generic under arbitrarily small C^1 perturbations. Then for any compact interval $[0, T]$, the number of crossings $N_{\mathrm{cross}}(0, T)$ is finite and obeys

$$N_{\mathrm{cross}}(0,T) \;\leq\; \frac{T}{\tau_{\min}}, \qquad \tau_{\min} \;=\; \frac{1}{L_F} \log\Bigl(1 + \frac{L_F\, r_{\mathrm{tube}}}{\alpha}\Bigr) \;\geq\; \frac{r_{\mathrm{tube}}}{\alpha + L_F r_{\mathrm{tube}}}\,.$$

Proof sketch. Within a tubular neighborhood of radius r_{tube} around Σ, let $d(t)$ be the signed normal distance to Σ (positive on one side). Along a Filippov solution, $\dot{d}(t) = \langle \dot{x}(t), n\rangle + \underbrace{\langle x - \pi_\Sigma(x), \nabla n\rangle}_{\text{curvature}}$, with the curvature term bounded by $L_F d(t)$ due to one-sided Lipschitzness and smoothness of n. Strict transversality gives $\dot{d} \geq \alpha - L_F d$, whose comparison solution yields the time to move from $-r_{\text{tube}}$ to $+r_{\text{tube}} \geq \tau_{\min}$; hence at most $T/\tau_{\min}$ crossings on $[0, T]$. □

Uses: (C1) r_{tube}; (C3) L_F and $\tau_{\min}$.

1.3 Bigraded exterior calculus on $T\Gamma \oplus TG$

On a stratum $M_i \subset G$ the product $\Gamma \times M_i$ carries classical multivector calculus. Set $\Lambda^{p,q}(\Gamma, M_i) := \Lambda^p(T\Gamma) \hat{\otimes} \Lambda^q(TM_i)$. Operations (mixed wedge $\wedge$, interior contraction ι, Schouten–Nijenhuis bracket $[\cdot,\cdot]_{SN}$) are standard and understood stratumwise.

1.4 Hybrid Calculus Axioms and geometry-scaled defect

Lemma 3 (Hybrid Calculus Axioms)**.** [Alg] *Let fields/forms be specified stratumwise with frontier-transversal compatibility (Def. 2). Along any Filippov trajectory x_t, the classical identities hold a.e. inside strata: graded-commutativity, contraction antiderivation, and bracket derivation. Jacobi holds up to a frontier defect Δ_{fr} with*

$$\int_0^T \|\Delta_{\text{fr}}(x_t)\|\, dt \;\leq\; C_{\text{fr}}(U, T)\, N_{\text{cross}}(0, T),$$

where C_{fr} depends on uniform bounds on fields and their first derivatives near frontiers and one-sided Lipschitz moduli.

Proof sketch. Test Jacobi on a smooth partition subordinate to a frontier tube. The only non-canceling terms are distributional fluxes supported on Σ, controlled by jumps of contractions across Σ; these are bounded by uniform field norms and Lipschitz moduli, yielding the stated L^1 estimate per crossing. □

Uses: (C2) $S_{\max}, G_{\max}$; (C3) one-sided Lipschitz L_F.

Lemma 4 (Jacobi defect with explicit geometry/angle scaling)**.** [Alg] *Under the hypotheses of Lemma 3 with frontier curvature $\kappa_{\max}$, tube radius r_{tube}, and minimal nonzero incidence angle $\theta_{\min}$, the per-crossing defect obeys*

$$\int_0^T \|\Delta_{\text{fr}}(x_t)\|\, dt \;\leq\; C_{\text{fr}} \cdot N_{\text{cross}}(0, T), \quad C_{\text{fr}} \;\leq\; (\|S\|_\infty + \|\nabla S\|_\infty)\left(1 + c_1\, \kappa_{\max} r_{\text{tube}}\right) \frac{c_2}{\sin\theta_{\min}},$$

with absolute constants c_1, c_2 independent of units, uniformly on compact sublevel sets.

Proof sketch. The tube interaction time scales like $r_{\text{tube}}/\sin\theta$; curvature perturbs n by $O(\kappa_{\max} r_{\text{tube}})$. The distributional commutator across Σ is controlled by jumps in contractions bounded by $\|S\|_\infty + \|\nabla S\|_\infty$, yielding the product form. □

Uses: (C1) $\kappa_{\max}, \theta_{\min}, r_{\text{tube}}$; (C2) $S_{\max}, G_{\max}$; (C3) $\tau_{\min}$.

Proposition 1 (Frontier-compatibility of derivations)**.** [Alg] *If derivations D_i agree in the frontier-transversal sense, their induced actions on $\Lambda^{\bullet,\bullet}$ satisfy the axioms with the same defect bound as in Lemma 3.*

Uses: (C1) frontier tubes; (C2) $S_{\max}, G_{\max}$; (C3) one-sided Lipschitz.

Numerical API (Frontier calculus: crossings, dwell time, geometry-scaled defect)

[Rep/Alg/Comp] This API detects frontier crossings, estimates a uniform dwell time $\widehat{\tau}_{\min}$ and a per-crossing Jacobi-defect budget $\widehat{C}_{\mathrm{fr}}$, and applies the frontier *freeze* rule used later by control and estimation layers.

Inputs. (i) A signed-distance or implicit frontier model $\phi(x)$ on $X = \Gamma \times G$ (positive/negative on adjacent strata); (ii) time-stamped states/velocities $(x_t, \dot{x}_t)$; (iii) tubular radius r_{tube}; (iv) field bounds $S_{\max}, G_{\max}$ and geometric moduli $\kappa_{\max}, \theta_{\min}$ (cf. §); (v) optional discrete mesh for DEC.

Outputs. Crossing log with times and incidence estimates; $\widehat{\tau}_{\min}$; $\widehat{C}_{\mathrm{fr}}$; freeze intervals; optional DEC error audit.

```python
# == 2.1 Frontier tube & normals ================================================
def build_frontier_tube(phi, r_tube):
    # Returns callable: signed distance d(x), unit normal n(x) on |phi(x)|<=r_tube
    def d(x): return phi(x)                      # assume phi ~ signed dist near frontier
    def n(x):                                    # robust normal from gradient
        g = grad_phi(x); return g / max(norm(g), 1e-12)
    return d, n

# == 2.2 Crossing detection & incidence metrics ================================
def detect_crossings(times, x, d, n, theta_min):
    # Returns list of crossings with normal component v_n and incidence angle
    crossings = []
    for k in range(1, len(times)):
        if d(x[k-1]) * d(x[k]) <= 0:          # sign change (robustify with hysteresis)
            v = (x[k] - x[k-1]) / (times[k]-times[k-1])
            v_n = abs(dot(v, n(x[k])))
            theta = max(theta_min, arccos( min(1.0, max(-1.0, dot(normalize(v), n(x[k]))))) )
            crossings.append({"t": times[k], "v_n": v_n, "theta": theta})
    return crossings

# == 2.3 Dwell time estimator (No-Zeno proxy) ==================================
def estimate_dwell_time(crossings, horizon_T):
    # use nearest-neighbor spacing between crossings
    if len(crossings) < 2: return {"tau_min_hat": horizon_T, "N_cross": len(crossings)}
    gaps = [crossings[i]["t"] - crossings[i-1]["t"] for i in range(1, len(crossings))]
    tau_hat = max(min(gaps), 0.0)
    return {"tau_min_hat": tau_hat, "N_cross": len(crossings), "gaps": gaps}

# == 2.4 Geometry-scaled defect budget =========================================
def estimate_C_fr(S_max, G_max, kappa_max, theta_vals):
    # heuristic upper bound: curvature + shallow incidence amplify interaction time
    inv_sin = [1.0/max(1e-3, sin(theta)) for theta in theta_vals]   # ~ 1/sin(theta)
    amp = max(inv_sin) if inv_sin else 1.0
    C0 = c_calib * (S_max + G_max) * (1.0 + kappa_max) * amp        # c_calib via unit tests
    return {"C_fr_hat": C0}

# == 2.5 Freeze rule scheduling ================================================
def schedule_freeze(crossings, tau_min_hat):
    # freeze on [t_k - eps, t_k + eps], with eps chosen <= tau_min_hat/2
    eps = 0.5 * tau_min_hat
```

```
    return [{"t0": c["t"] - eps, "t1": c["t"] + eps} for c in crossings]

# == 2.6 Optional DEC audit ==========================================================
def dec_consistency(mesh_h, k_order, N_cross):
    # stratumwise O(h^k) + frontier O(h^k) per crossing
    err = C_dec * (mesh_h**k_order) * (1 + N_cross)
    return {"dec_err_bound": err, "k": k_order, "h": mesh_h}

# == 2.7 Orchestration ===============================================================
def frontier_runtime(times, x, phi, r_tube, S_max, G_max, kappa_max, theta_min, mesh=None):
    d, n = build_frontier_tube(phi, r_tube)
    crossings = detect_crossings(times, x, d, n, theta_min)
    dwell = estimate_dwell_time(crossings, horizon_T=times[-1]-times[0])
    Cfr = estimate_C_fr(S_max, G_max, kappa_max, [c["theta"] for c in crossings])
    freezes = schedule_freeze(crossings, dwell["tau_min_hat"])
    audit = dec_consistency(mesh.h, mesh.k, dwell["N_cross"]) if mesh else {}
    log = {"N_cross": dwell["N_cross"], "tau_min_hat": dwell["tau_min_hat"],
           "C_fr_hat": Cfr["C_fr_hat"], "freezes": freezes, **audit}
    return log
```

Acceptance checks. (i) Logs report N_{cross}, $\widehat{\tau}_{\min}$, and $\widehat{C}_{\mathrm{fr}}$; (ii) freeze intervals are non-overlapping and contained in the tube; (iii) optional DEC audit prints $O(h^k) + N_{\mathrm{cross}} O(h^k)$.
Uses: r_{tube}, L_{F}, $\theta_{\min}$, $\kappa_{\max}$, $S_{\max}$, $G_{\max}$.

1.5 DEC interoperation with frontier term

Let $\mathcal{T}_h$ be a shape-regular simplicial (or cubical) mesh on each M_i with mesh size h, and let W_h denote Whitney forms. Denote by $\mathsf{W}_h : \Omega^\bullet \to \Omega^\bullet_h$ the Whitney projection and by R_h its de Rham inverse (integration on chains). Assume crossings satisfy Lemma 2 and fields are C^{k+1} on strata.

Proposition 2 (DEC consistency with frontier term and cut-cell quadrature)**.** [Alg/Comp] *Assume either* (a) *fitted meshes with uniform angle bounds on cut cells, or* (b) *unfitted meshes with a ghost-penalty/aggregate stabilization on the tube. For* $0 \le r \le k$ *and each stratum* M_i*,*

$$\|\mathsf{W}_h(\omega \wedge \eta) - \mathsf{W}_h\omega \wedge_h \mathsf{W}_h\eta\|_{L^2(M_i)} \le C\, h^{k-r} \|\omega\|_{H^k} \|\eta\|_{H^k},$$

and similarly for brackets. Along any Filippov trajectory x_t *on* $[0,T]$*,*

$$\int_0^T \|\Delta_{\mathrm{fr},h}(x_t)\|\, dt \;\le\; C\, h^{k-r}\, T \;+\; \underbrace{C_{\mathrm{quad}}\, h^q\, T}_{\text{cut-cell quadrature, } q \ge k} \;+\; C_{\mathrm{fr}}\, N_{\mathrm{cross}}(0,T).$$

Thus the total defect is $O(h^{k-r}T + h^q T) + O(C_{\mathrm{fr}}\, T/\tau_{\min})$.
Uses: Shape regularity or ghost-penalty on cut cells; Lem. 2; Lem. 4.

Acceptance checks for Section 1.

1. All later bounds that invoke frontier disturbance (e.g., ISS in §??) reference both $\tau_{\min}$ (Lemma 2) and C_{fr} (Lemma 4).
2. Any occurrence of N_{cross} appears only inside expressions dominated by $T/\tau_{\min}$ on finite

horizons.

3. The DEC bridge exposes explicit h-rates and a frontier contribution that scales as $C_{\mathrm{fr}}\,T/\tau_{\min}$.

2 Strategic Series and Cross-Cumulants

We formalize strategic fields on Γ and G, define scalar evaluation functionals, and construct mixed cumulants under invariant or windowed laws. New in this draft: (i) measurability and L^2 conditions for $T_{u,v}$ (Lemma 5); (ii) a corrected factorization statement that separates the *global* "separating dictionary" case from the *finite-K* practical case (Proposition 3); (iii) a quantitative windowed stability bound (Lemma 6).

2.1 Strategic series and evaluation functionals

On each stratum $M_i \subset G$, take the bigraded series $\hat{S} = \sum_{p,q} S^{(p,q)}$ with

$$S^{(p,q)} \in \Gamma\big(\Lambda^p(T\Gamma)\,\hat{\otimes}\,\Lambda^q(TM_i)\big).$$

For covectors $u \in \Lambda^p(T^*\Gamma)$ and $v \in \Lambda^q(T^*G)$ define the scalar *evaluation functional*

$$T_{u,v}(x) \;:=\; \iota_{u\otimes v}\,\hat{S}(x)\,, \qquad x = (\gamma, g) \in \Gamma \times M_i,$$

interpreted stratumwise (Lemma 3 ensures algebraic well-posedness a.e. along Filippov trajectories).

2.2 Existence of laws and windowing

Under Assumption 1, standard Lyapunov/tightness arguments deliver invariant laws μ for $(X_t)_{t\ge 0}$ when the drift/diffusion contracts on average; otherwise we use *windowed laws* on $[t-\tau, t]$:

$$\mu_{[t-\tau,t]} \;:=\; \frac{1}{\tau}\int_{t-\tau}^{t} \delta_{X_s}\, ds.$$

All statements below hold either for μ or uniformly in t for $\mu_{[t-\tau,t]}$ when τ exceeds the local mixing time.

Lemma 5 (Measurability and L^2 of evaluation functionals)**.** [Stoch/Alg] *Suppose on each stratum M_i: (i) $S^{(p,q)}$ is Borel and locally square-integrable; (ii) (u, v) have at most polynomial growth in charts; (iii) crossings satisfy Lemma 2. Then $T_{u,v}$ is Borel on $X = \Gamma \times G$ and $T_{u,v} \in L^2(\mu)$ (or $L^2(\mu_{[t-\tau,t]})$).*

Proof sketch. Carathéodory measurability of contractions in local charts and the frontier-transversal compatibility (Def. 2) yield Borel measurability; local L^2 on strata and finite crossing count ensure global L^2 under μ or window averages. □

Uses: (C2) $S_{\max}$ bounds; (C3) No-Zeno $\tau_{\min}$; Rep charts.

2.3 Mixed cumulants and factorization

Let $\mathcal{T}_K = \{T_{p,q}^{(m)}\}_{1 \le p+q \le K}$ be a centered, whitened finite dictionary (Section 4). For a finite list $(Z_1, \dots, Z_m)$ drawn from $\mathcal{T}_K$, define the (alternating) joint cumulant $\kappa(Z_1, \dots, Z_m)$ via the usual moment–cumulant Möbius inversion; we write $\kappa_{p,q}$ for degree (p, q) cumulants.

Proposition 3 (Factorization via vanishing mixed cumulants)**.** [Stoch/Inf] *Consider γ-only statistics and G-only statistics drawn from (possibly infinite) dictionaries.*

(i) Global, separating dictionary. *If the dictionary separates points and the cumulant generating function is analytic near 0, then*

$$\gamma \;\perp\; G \;\big|\; Y \qquad \Longleftrightarrow \qquad \kappa_{p,q} = 0 \;\; \textit{for all } p, q \ge 1.$$

(ii) Finite-K implication with quantitative residual. *Fix total order $K \in \{2, 3, \dots\}$ and a centered, whitened finite dictionary $\mathcal{T}_K$. If $\kappa_{p,q} = 0$ for all $p, q \ge 1$ with $p + q \le K$, then for any polynomial functional F of $\mathcal{T}_K$ with total degree $\le K$,*

$$\Big| \mathbb{E}[F(\gamma, G)] \;-\; \mathbb{E}[F_\Gamma(\gamma)]\, \mathbb{E}[F_G(G)] \Big| \;\le\; C_K(F) \sum_{p+q \ge K+1} \|\kappa_{p,q}\|,$$

where $C_K(F)$ depends only on K and the coefficients of F in the whitened basis. In particular, the log-moment generating function satisfies

$$\psi(\theta_\Gamma, \theta_G) \;=\; \psi_\Gamma(\theta_\Gamma) + \psi_G(\theta_G) \;+\; R_{K+1}(\theta), \qquad |R_{K+1}(\theta)| \;\le\; C'_K \sum_{p+q \ge K+1} \|\kappa_{p,q}\|\, \|\theta\|^{p+q},$$

for $\|\theta\|$ within the EF radius of Section 4.

Proof sketch. (i) follows from the Lévy–Khintchine/moment–cumulant expansion with a separating dictionary. (ii) truncates the series at total degree K; the remainder aggregates neglected mixed cumulants with combinatorial coefficients (Bell polynomials) that are bounded by $C_K(F)$ after whitening. The EF radius control comes from Theorem 2/Remark **??**. □

Uses: (C5) whitened $F = \mathrm{FIM}(0) \succ 0$, EF radius $r \le \epsilon / L_\psi$.

Interpretation (cross-terms). Cross-terms in $\hat{S}$ map to *mixed* cumulants $\kappa_{p,q}$: nonzero $\kappa_{p,q}$ for $p, q \ge 1$ are the irreducible coupling certificates between decision flow on Γ and structure updates on G. Proposition 3 formalizes the "no cross-term ⇒ factorization up to order K" claim used in closure policies.

2.4 Windowed stability under drift

Let L be the W_1 drift speed of the ground-truth law across time (Section 3). For windowed cumulants computed on $[t - \tau, t]$ and $[s - \tau, s]$ with $|t - s| \ll \tau$:

Lemma 6 (Windowed stability of mixed cumulants)**.** [Stoch] *If each statistic in $\mathcal{T}_K$ is L_T-Lipschitz in d_X, then for $p + q \le K$,*

$$\Big| \kappa_{p,q}^{(t,\tau)} - \kappa_{p,q}^{(s,\tau)} \Big| \;\le\; C_{p,q}\, L_T\, \frac{|t - s|}{\tau} \;+\; o_\tau(1),$$

uniformly on compact sublevel sets. The constant $C_{p,q}$ depends only on p, q and the dictionary normalization.

Proof sketch. Write cumulants as multilinear polynomials in moments; bound each moment difference by the Kantorovich–Rubinstein dual with L_T and the window averaging factor $|t-s|/\tau$, then propagate through the multilinear map. □

Uses: (C4) τ; Lipschitz statistics; window laws $\mu_{[t-\tau,t]}$.

Acceptance checks for Section 2.

1. Proposition 3 *(ii)* is the only statement invoked when working with a finite dictionary; any "⇔" claim elsewhere references *(i)* with a separating dictionary.
2. All L^2 and measurability claims for $T_{u,v}$ cite Lemma 5.
3. Any time-variation bounds for windowed cumulants cite Lemma 6 with explicit dependence on $|t-s|/\tau$.

Numerical API (Strategic series, windowed cumulants, finite-K tests)

[Stoch/Inf/Comp] This API constructs test functionals $T_{u,v}$, performs invariant whitening, computes windowed cumulants up to order $K \le 3$ by streaming U-statistics, and runs a finite-K factorization test (â‡', not â‡").

Inputs. (i) Centered raw features/features-of-forms for letters $a_{p,q}^{(m)}$; (ii) symmetry group actions (frames on Γ, graph isomorphisms on G) for invariant whitening; (iii) window spec (τ, stride); (iv) target order $K \in \{2,3\}$.

Outputs. Per-window $\widehat{\mathbf{k}}_K$, FIM norm, HAC/BB hooks (from Section 9), and a BH/BY-controlled finite-K factorization report.

```
# == 3.1 Invariant whitening ==========================================================
def orbit_invariant_cov(features, group_actions, nsamp=128):
    # Monte Carlo over group actions to enforce symmetry-invariant inner product
    Sigma = 0
    for g in sample(group_actions, nsamp):
        Xg = apply_action(features, g)           # permute/rotate letters
        Sigma += cov(Xg)
    Sigma /= nsamp
    return Sigma

def whiten_dictionary(features, Sigma):
    W = inv(cholesky(Sigma))                    # Fisher/whitening map
    return features @ W.T, W                    # whitened letters, map

# == 3.2 Test functionals T_{u,v} ==============================================
def build_T_uv(letters, u, v):
    # u,v are contraction descriptors into bigraded series \hat S
    def T(x):
        # evaluate contractions using current \hat S and x=(\gamma,G)
        return contract(u, v, letters, x)
    return T

```

```
# == 3.3 Streaming cumulants (order <= 3) ===================================
def stream_cumulants(T_stream, K=3):
    # maintains running moments and converts to cumulants via M\"obius inversion
    state = init_u_stats(K)
    for Tt in T_stream: update_u_stats(state, Tt, K)
    return u_to_cumulants(state, K)             # dict of kappa_{p,q}^{(m)}

# == 3.4 Windowed pipeline ==================================================
def windowed_cumulants(data_stream, letters, Sigma, K, win_size, stride):
    T_wh, W = whiten_dictionary(letters, Sigma)
    windows = sliding_windows(data_stream, win_size, stride)
    out = []
    for w in windows:
        kappa = stream_cumulants(eval_T_stream(T_wh, w), K=K)
        kvec  = flatten_kappa(kappa)            # stack by (p,q,m)
        F     = fisher_at_P0(T_wh)              # = I if fully whitened
        fim_norm = sqrt(kvec.T @ inv(F) @ kvec)
        out.append({"window": w.id, "kappa": kappa, "kvec": kvec, "fim_norm": fim_norm})
    return out

# == 3.5 Finite-K factorization test (BH/BY) ================================
def finiteK_factorization_test(window_out, groups, q=0.1, method="BH"):
    # H0: selected entries of mixed cumulants == 0  (vanish up to K)
    pvals = {}
    for win in window_out:
        pvals[win["window"]] = test_zero_kappa(win["kvec"])  # HAC/BB from section 9
    # group by bidegree/stratum per section 9
    rejections = {}
    for gid, P in groups.items():
        rejections[gid] = fdr_bh(P, q) if method=="BH" else fdr_by(P, q)
    return {"pvals": pvals, "reject": rejections}

# == 3.6 Diagnostics ========================================================
def mixing_diagnostic(series, max_lag=64):
    ac = autocorr(series, max_lag)
    return {"ac_tail": ac[-5:], "mix_flag": (abs(ac[max_lag])<0.1)}
```

Acceptance checks. (i) Whitening uses orbit-averaged Σ (equivariant); (ii) each window reports $\|\widehat{\mathbf{k}}_K\|_{\mathrm{FIM}^{-1}}$; (iii) finite-$K$ test declares only the $\Rightarrow$ direction (vanishing up to K $\Rightarrow$ factorization up to K), with BH/BY and group structure as in Section 9; (iv) mixing diagnostic is logged for each window.

Uses: Invariant whitening (Assumption **??**); Fisher F; Section 9 HAC/BB/FDR.

3 Hierarchical Closure: BBGKY Analogy and DRO

We pose closure as a Wasserstein–KL distributionally robust optimization (DRO) and derive a tractable soft-transport dual. New here: (i) a stratified *reference measure* λ that is *refinement invariant* (Definition 4, Proposition **??**); (ii) measurability and stratum-Lipschitz properties of the soft potential φ_η (Lemma 8); (iii) differentiability of $\mathcal{R}_\rho$ in (θ, η) with an envelope/KKT system and explicit gradients (Proposition 5); (iv) an *adaptive radius* policy with guarantees that expose

finite-sample W_1 rates via an effective transport dimension d_{eff} (Definition 5, Theorem 1).

3.1 DRO objective and soft-transport dual

Given a whitened statistic dictionary up to order K (Section 4), we fit θ by

$$\min_{\theta\in\Theta} \underbrace{\sup_{Q:\ W_1(Q,\hat{P})\le\rho} D_{\mathrm{KL}}(Q\|P_\theta)}_{:=\ \mathcal{R}_\rho(\theta)} + \lambda_{\text{hopf}} \left\|(\mathrm{id}-\Pi_{\le K})\,\hat{S}\right\|^2. \tag{1}$$

Here W_1 is computed with a ground metric d_X on $X=\Gamma\times G$, and the second term penalizes Hopf-residual energy (Section **??**).

Definition 4 (Stratified reference with continuous+atomic split)**.** Write $\lambda=\lambda_{\text{cont}}+\lambda_{\text{atom}}$ with

$$\lambda_{\text{cont}} := \sum_{i\in I} w_i\,\mathcal{H}^{\dim M_i}\lfloor_{M_i}, \qquad \lambda_{\text{atom}} := \sum_{j\in J} w'_j\,\delta_{D_j},$$

normalized on compacta and refinement-invariant (Prop. 4). For any model P_θ, write $P_\theta = P_\theta^{\text{cont}} + P_\theta^{\text{atom}}$ with $P_\theta^{\text{cont}} \ll \lambda_{\text{cont}}$ and $P_\theta^{\text{atom}} \ll \lambda_{\text{atom}}$.

Proposition 4 (Refinement-invariant stratified reference measure)**.** [Stoch/Clos] *Let $G=\bigsqcup_i M_i$ be Whitney-stratified. Choose weights $c_i \propto \mathcal{H}^{\dim M_i}(M_i\cap K)$ on compacta K (Hausdorff volume of top-dimensional pieces restricted to K) and assign atom weights c'_j continuously under refinements so that $\sum_i c_i + \sum_j c'_j = 1$ on K. If two stratifications admit a common refinement, the induced measures λ are mutually absolutely continuous and the inner value $\mathcal{R}_\rho(\theta)$ is identical for both.*

Proof sketch. Use Carathéodory's construction on the stratified σ-algebra and note that Hausdorff measures agree up to equivalence across common refinements on each piece; atomic mass continuity prevents mass creation under partitioning. The dual integral defining φ_η is invariant under equivalent λ. □

Uses: Whitney stratification; Hausdorff measures; Carathéodory extension.

Lemma 7 (Differentiability and KKT envelope)**.** [Stoch/Clos/Inf] *Assume: (i) $dP_\theta/d\lambda$ is C^1 in θ on each $\Gamma\times M_i$; (ii) there exists an $L^1(\lambda)$ envelope G with $\sup_{\|\theta\|\le r,\ \eta\in[0,\eta_{\max}]} \left\|\partial_\theta \log \frac{dP_\theta}{d\lambda}\right\| \le G$; and (iii) d_X is continuous. Then*

$$\nabla_\theta \mathcal{R}_\rho(\theta) \;=\; \nabla_\theta \mathbb{E}_{\hat{P}}[\varphi_{\eta^\star}(\cdot;\theta)] \;=\; \mathbb{E}_{\hat{P}}\left[\frac{\int e^{-\eta^\star d_X(z,\cdot)}\,\partial_\theta \frac{dP_\theta}{d\lambda}(z)\,d\lambda(z)}{\int e^{-\eta^\star d_X(z,\cdot)}\,\frac{dP_\theta}{d\lambda}(z)\,d\lambda(z)}\right],$$

and the KKT system holds with $\eta^\star\ge 0$ solving $\rho = \mathbb{E}_{\hat{P}}[\partial_\eta \varphi_\eta]$.

Proof sketch. Carathéodory measurability (Lemma 8) + dominated convergence (envelope) justify interchanging ∇_θ and the integral; apply the envelope theorem for the inner infimum in η. □

Uses: Envelope $G\in L^1$; Lem. 8.

Remark 2 (Finite-sample W_1 drift rates and δ in Thm. 1). *[Stoch/Clos]* With n points per window and effective transport dimension d_{eff}, $|\hat{L}_t - L| = O_{\mathbb{P}}(n^{-1/d_{\text{eff}}})$ in the worst case, and improves to $O_{\mathbb{P}}(n^{-1/2})$ under bounded transport dimension / sub-Gaussian dictionary tails. Choose δ in Theorem 1 accordingly to set (c,τ) pragmatically.

Uses: Window size n; dictionary tails/transport-dimension assumption.

With $P_\theta \ll \lambda$ and continuous d_X, the inner sup admits the Donsker–Varadhan/Kantorovich dual

$$\mathcal{R}_\rho(\theta) \;=\; \inf_{\eta\ge 0}\Big\{\eta\rho + \mathbb{E}_{\hat P}[\varphi_\eta(\cdot;\theta)]\Big\}, \quad \varphi_\eta(x;\theta) \;:=\; \log\int \exp\big(-\eta d_X(z,x)\big)\,\frac{dP_\theta}{d\lambda}(z)\,d\lambda(z).$$

Lemma 8 (Measurability and stratum-Lipschitz of φ_η)**.** [Stoch/Clos] *If d_X is continuous and $z \mapsto \frac{dP_\theta}{d\lambda}(z)$ is stratumwise continuous, then $x \mapsto \varphi_\eta(x;\theta)$ is Borel on X and locally Lipschitz on each $\Gamma \times M_i$.*

Proof sketch. Carathéodory's criterion applies to $(x,z) \mapsto e^{-\eta d_X(z,x)}\,dP_\theta/d\lambda(z)$; local dominated convergence on strata yields stratum-Lipschitzness of φ_η. □

Uses: (C1) stratum charts; continuity of d_X; stratumwise continuity of $dP_\theta/d\lambda$.

3.2 Differentiability and KKT system

Proposition 5 (Differentiability and KKT for the soft-transport dual)**.** [Clos/Inf] *Suppose for some $r > 0$, $\eta \in [0, \eta_{\max}]$, and all $\|\theta\| \le r$: (i) $dP_\theta/d\lambda$ is C^1 stratumwise with an $L^1(\lambda)$ envelope uniform in (θ,η); (ii) $\varphi_\eta(x;\theta)$ from Lemma 8 is $\hat P$-integrable. Then $\mathcal{R}_\rho(\theta)$ is C^1 on $\{\|\theta\| \le r\}$, and the inner minimization over $\eta \ge 0$ admits KKT conditions with*

$$\partial_\eta\Big(\eta\rho + \mathbb{E}_{\hat P}[\varphi_\eta]\Big)\Big|_{\eta=\eta^\star} \;=\; 0 \quad\Longleftrightarrow\quad \rho \;=\; \mathbb{E}_{\hat P}\Big[\underbrace{\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}[d_X(Z,x)]}_{:=\,\bar d_\eta(x;\theta)}\Big],$$

where $Q^x_{\eta,\theta}$ is the Gibbs kernel on X with density

$$\frac{dQ^x_{\eta,\theta}}{d\lambda}(z) \;\propto\; e^{-\eta d_X(z,x)}\,\frac{dP_\theta}{d\lambda}(z).$$

Moreover,

$$\nabla_\theta \mathcal{R}_\rho(\theta) \;=\; \mathbb{E}_{\hat P}\Big[\;\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}\big[\;\nabla_\theta \log(dP_\theta/d\lambda)(Z)\;\big]\;\Big].$$

If $\eta \mapsto \mathbb{E}_{\hat P}[\varphi_\eta]$ is strictly convex (e.g. when d_X is non-degenerate on supports), then $\eta^\star$ is unique and the envelope theorem applies.

Proof sketch. Differentiate φ_η under the integral using the uniform L^1 envelope; apply Danskin's theorem/envelope theorem at the η-minimizer. The ∂_η identity is the well-known soft-transport moment. The θ-gradient follows from the log-derivative trick inside the Gibbs kernel. □

Uses: Dominated convergence envelope; (C5) EF radius context; (C4) window integrability.

Corollary 1 (KKT and gradient under $\lambda_{\text{cont}} + \lambda_{\text{atom}}$)**.** *With Definition 4, the soft-transport dual reads $\mathcal{R}_\rho(\theta) = \inf_{\eta\ge 0}\{\eta\rho + \mathbb{E}_{\hat P}[\varphi_\eta]\}$ with*

$$\varphi_\eta(x;\theta) = \log\Big(\int e^{-\eta d_X(z,x)}\,\frac{dP^{\text{cont}}_\theta}{d\lambda_{\text{cont}}}(z)\,d\lambda_{\text{cont}}(z) \;+\; \sum_{D\in\text{Atoms}} e^{-\eta d_X(D,x)}\,\frac{dP^{\text{atom}}_\theta}{d\lambda_{\text{atom}}}(D)\Big).$$

The KKT condition is $\rho = \mathbb{E}_{\hat P}[\bar d_\eta(x;\theta)]$ with $\bar d_\eta$ the $Q^x_{\eta,\theta}$-expectation over the product *measure, and*

$$\nabla_\theta \mathcal{R}_\rho(\theta) = \mathbb{E}_{\hat P}\Big[\mathbb{E}_{Z\sim Q^x_{\eta^\star,\theta}}[\nabla_\theta \log(dP^{\text{cont}}_\theta/d\lambda_{\text{cont}})(Z)]\Big] \;+\; \sum_D \hat P(\{x\})\,\pi_{\eta^\star,\theta}(D \mid x)\,\nabla_\theta \log\frac{dP^{\text{atom}}_\theta}{d\lambda_{\text{atom}}}(D).$$

Lemma 9 (Uniqueness of $\eta^\star$ by strict decrease)**.** *If $d_X(\cdot, x)$ is not λ-a.e. constant for each x (GKYP-type nondegeneracy), then for fixed (θ, x)*

$$g_x'(\eta) = -\operatorname{Var}_{Q^x_{\eta,\theta}}[d_X(Z, x)] \;<\; 0.$$

Hence $g(\eta) := \mathbb{E}_{\hat{P}}[g_x(\eta)]$ is strictly decreasing and the KKT root $\eta^\star$ is unique.

Proposition 6 (DRO-metric sensitivity)**.** *Let $d_X^\epsilon = d_X + \epsilon\, \delta d$ with δd Lipschitz and $\theta^*(\epsilon)$ be the minimizer of (1). Under the IFT assumptions for the outer problem, $\partial_\epsilon \theta^*(0) = -\left[\nabla^2_\theta \mathcal{R}_\rho(\theta^*)\right]^{-1} \nabla_\theta \mathbb{E}_{\hat{P}}[\partial_\epsilon \varphi_{\eta^*}(\cdot; \theta^*)]$, with $\partial_\epsilon \varphi_{\eta^*}(x; \theta) = -\eta^*\, \mathbb{E}_{Q^x_{\eta^*,\theta}}[\delta d(Z, x)]$. Report: log $\|\partial_\epsilon \theta^*(0)\|$ as "DRO-metric sensitivity".*

Numerical note. Compute φ_η with log-sum-exp stabilization; sample from $Q^x_{\eta^\star,\theta}$ via importance sampling or short-run MCMC confined to a stratum (accept/reject at frontiers).

3.3 Adaptive radius policy: finite-sample rates

Definition 5 (Adaptive radius policy)**.** *[Clos/Stoch]* For window size τ with n effective observations and transport dimension d_{eff} for (X, d_X) on the window, define a pilot drift estimate

$$\hat{L}_t \;:=\; \tfrac{1}{\tau}\, W_1(\hat{P}_{[t-\tau,t]}, \hat{P}_{[t-2\tau,t-\tau]}), \qquad \rho_t \;:=\; c\, \hat{L}_t\, \tau \quad (c \geq 1).$$

Uses: (C4) τ, ρ; (C1) metric measurability.

Theorem 1 (Containment and regret under finite-sample W_1 rates)**.** [Clos] *Let $P_{[t-\tau,t]}$ and $P_{[t-2\tau,t-\tau]}$ be the true window laws with W_1 drift speed L. Suppose $\hat{P}$'s are empirical with n effective points and*

$$|\hat{L}_t - L| \;\leq\; \delta_{n,d_{\text{eff}}} \quad \textit{with} \quad \delta_{n,d_{\text{eff}}} = \begin{cases} C\, n^{-1/d_{\text{eff}}} & \textit{(worst-case, no moment/low-dim assumptions)}, \\ C'\, n^{-1/2} & \textit{(under bounded transport dimension / sub-Gaussian tails)}. \end{cases}$$

Then with $\rho_t = c\, \hat{L}_t\, \tau$:

1. *(*Containment*) $P_{[t-\tau,t]} \in \mathcal{W}_1(\hat{P}_{[t-\tau,t]}, \rho_t)$ with probability $\geq 1 - \alpha$ for a choice of $c = c(\alpha)$ depending on the tail bound of $\delta_{n,d_{\text{eff}}}$.*
2. *(*Regret drift term*) The drift contribution in $D_{\mathrm{KL}}(P \| P_{\widehat{\theta}})$ scales as*

$$\text{Regret}_{\text{drift}} \;\leq\; C_d(L + \delta_{n,d_{\text{eff}}})\, \tau.$$

Proof sketch. Triangle inequality for W_1 and concentration of empirical W_1 around truth give the containment for appropriate c. The regret bound follows from the dual representation with Lemma 8 (Lipschitz in x) and Proposition 5 to control the variation of $\varphi_{\eta^\star}$ across adjacent windows. □

Uses: (C4) τ; Lipschitz constants from Lemma 8; envelope from Proposition 5.

3.4 Error decomposition (updated)

With the adaptive radius, the standard decomposition becomes

$$D_{\mathrm{KL}}(P \| P_{\widehat{\theta}}) \;\leq\; C_a \|(\mathrm{id} - \Pi_{\leq K})\hat{S}\|^2 \;+\; C_e n^{-1/2} \;+\; C_d(L + \delta_{n,d_{\text{eff}}})\, \tau,$$

where the last term replaces the purely qualitative drift piece and is reported per window in logs together with $\eta^\star$.

Acceptance checks for Section 3.

1. Every use of the soft-transport dual references Definition 4 and Lemma 8; claims about invariance under re-stratification cite Proposition **??**.
2. KKT/gradient statements cite Proposition 5 and list the envelope hypotheses; implementations log $\eta^\star$.
3. The adaptive radius and regret bounds cite Theorem 1 with explicit n and d_{eff} dependence; runs log $\widehat{L}_t$, ρ_t, $\eta^\star$.

Numerical API (soft-transport dual, gradients, and logging)

```
# Ground types (conceptual)
Metric dX(x,z)                        # ground metric on X = Gamma x G
Model  P_theta                        # exponential-family or other density; exposes:
                                      #   logpdf(z; theta), score(z; theta) = grad_theta logpdf
Measure lambda                        # refinement-invariant reference (Def.~4)
Sampler draw_from_lambda(stratum, m)  # draws z~lambda restricted to a stratum (or global)
Projector Pi_leqK                     # Hopf projector used in residual penalty
Logger log_json(dict)                 # appends a JSON line to the run log

# ---------- Core primitives (stable numerics) ----------
function varphi_eta(x_batch, theta, eta, dX, P_theta, lambda, strata_policy):
    # returns: vector phi[i] = varphi_eta(x_i; theta)
    for each x in x_batch:
        S = 0.0            # log-sum-exp accumulator (log domain)
        w_sum = 0.0
        # importance base: sample M support points from lambda (same stratum as x by default)
        Z = draw_from_lambda(strata_policy(x), M)
        # compute log-weights with log-sum-exp stabilization
        for z in Z:
            \ell = -eta * dX(z, x) + logpdf_P_over_lambda(z; theta, P_theta, lambda)
            # standard log-sum-exp trick
            if S == 0.0: S = \ell
            else:
                if \ell > S: w_sum = exp(S - \ell) * w_sum + 1.0; S = \ell
                else:        w_sum = w_sum + exp(\ell - S)
        phi[x] = S + log(w_sum) - log(M)        # Monte Carlo approx. to log \int ...
    return phi

function soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy):
    # returns a closure that can sample/reweight from Q_{eta,theta}^x via importance weights on lambda
    Z = draw_from_lambda(strata_policy(x), M)
    L = [ -eta*dX(z,x) + logpdf_P_over_lambda(z; theta, P_theta, lambda) for z in Z ]
    S = max(L); w = [ exp(\ell - S) for \ell in L ]; w = w / sum(w)    # normalized weights
    return (Z, w)   # usage: E_{Q}[f] \approx sum_j w[j] * f(Z[j])

# ---------- Inner solver for \eta^\star ----------
function eta_star(rho, x_batch, theta, dX, P_theta, lambda, strata_policy,
                  bracket=[1e-6, 1e3], tol=1e-5, M=512):
    # solve g(eta) = E_x E_{Q_{eta,theta}^x}[ dX(Z,x) ] - rho = 0
```

```
    function g(eta):
        acc = 0.0
        for x in x_batch:
            (Z,w) = soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy)
            acc += sum_j w[j] * dX(Z[j], x)
        return acc / len(x_batch) - rho
    # bracket monotone root (g is decreasing in eta under mild conditions)
    a,b = bracket
    if g(a) < 0: return a
    if g(b) > 0: increase b geometrically until g(b) < 0 or b > 1e6
    # Brent or bisection (robust)
    while |b-a| > tol*(1+min(|a|,|b|)):
        c = 0.5*(a+b)
        if g(c) > 0: a = c else: b = c
    return 0.5*(a+b)

# ---------- Risk and gradient ----------
function R_rho(theta, x_batch, rho, dX, P_theta, lambda, strata_policy, M=1024):
    eta = eta_star(rho, x_batch, theta, dX, P_theta, lambda, strata_policy, M=M/2)
    phi = varphi_eta(x_batch, theta, eta, dX, P_theta, lambda, strata_policy)
    return eta*rho + mean(phi), eta

function grad_R_rho(theta, x_batch, eta, dX, P_theta, lambda, strata_policy, M=1024):
    # \nabla_\theta R_rho(\theta) = E_x E_{Q_{eta,theta}^x}[ score(Z; theta) ]
    g = 0 (shape like theta)
    for x in x_batch:
        (Z,w) = soft_transport_kernel(x, theta, eta, dX, P_theta, lambda, strata_policy)
        g += sum_j w[j] * score(Z[j]; theta)
    return g / len(x_batch)

# ---------- Adaptive radius and drift logging ----------
function adaptive_radius(window_curr, window_prev, c, dX, effective_n, d_eff_hint=None):
    # window_curr/prev: datasets; c >= 1
    \hat W1 = empirical_W1(window_curr, window_prev, dX)      # same metric used in dual
    \hat L  = \hat W1 / window_length_tau
    rho_t   = c * \hat L * window_length_tau
    # optional: rate hint for monitoring (not used in computation)
    if d_eff_hint is None: rate = C / sqrt(effective_n) else: rate = C / (effective_n)^(1/d_eff_hint)
    return rho_t, \hat L, \hat W1, rate

# ---------- One-call convenience wrapper per window ----------
function dro_step(theta, window_curr, window_prev, config):
    # 1) adaptive radius (logs drift)
    rho_t, \hat L, \hat W1, rate_hint = adaptive_radius(window_curr, window_prev,
                            config.c_radius, dX=config.dX, effective_n=config.n_eff,
                            d_eff_hint=config.d_eff)
    # 2) pick minibatch x_batch from window_curr (or full window)
    x_batch = minibatch(window_curr, size=config.batch)
    # 3) compute risk and eta*
    R, eta = R_rho(theta, x_batch, rho_t, config.dX, config.P_theta, config.lambda, config.strata_polic
    # 4) gradient (for optimizer)
    g = grad_R_rho(theta, x_batch, eta, config.dX, config.P_theta, config.lambda, config.strata_policy,
    # 5) Hopf residual (for penalty term and audit)
    S_resid = norm( (Id - Pi_leqK) @ hatS(window_curr) )^2
```

```
    # 6) structured logging
    log_json({
        "window_t": config.t_id,
        "rho_t": rho_t,
        "L_hat": \hat L,
        "W1_hat": \hat W1,
        "eta_star": eta,
        "R_rho": R,
        "hopf_residual": S_resid,
        "n_eff": config.n_eff,
        "rate_hint": rate_hint,
        "M_mc": config.M_mc,
        "batch": config.batch
    })
    return R + config.lambda_hopf * S_resid, g
```

Recommended defaults and guards.

- *Bracketing for $\eta^\star$.* Start with $[10^{-6}, 10^3]$; expand the right endpoint by $\times 10$ if $g(b) > 0$. Enforce a hard cap (e.g., 10^8) and fall back to $\eta^\star = b$ if the bracket fails—log this.
- *Monte Carlo budget.* Use $M = 512$–2048 per x; reuse samples across nearby η during the 1-D solve (common-random-numbers variance reduction).
- *Stratum policy.* By default, draw Z from the *same* stratum as x; allow "cross-stratum" proposals only if the frontier tube flag is raised (then freeze Π and θ as in Section 6/Section 7).
- *Stability.* All integrals in φ_η and kernel weights use log-sum-exp. Clip η to avoid $\exp(-\eta d)$ underflow; report the clip in the log.
- *EF models.* For exponential families, `score(z;theta)` is available in closed form; cache $F = \mathrm{FIM}(0)$ for whitening and preconditioning. Use a Fisher preconditioner for the optimizer step.
- *Unit tests.* (i) At $\rho = 0$, verify $\eta^\star = 0$, $R_\rho(\theta) = \mathbb{E}[\log \frac{dP_\theta}{d\lambda}(X)]$. (ii) If $d_X \equiv 0$, confirm $\eta^\star$ arbitrary and R_ρ reduces to the log-normalizer term. (iii) Under independence (negative control), $\eta^\star \approx 0$ and the Hopf residual dominates the objective.

Minimal config stub (to mirror your Section 10 API).

```
config := \{
  dX: ground\_metric\_on\_X,\quad
  P\_theta: ef\_model,\quad
  lambda: stratified\_reference\_measure,\quad
  strata\_policy: same\_stratum\_by\_default,\quad
  c\_radius: 1.0,\quad
  t\_id: window\_index,\quad
  n\_eff: effective\_sample\_size,\quad
  d\_eff: None\ |\ estimated\_transport\_dimension,\quad
  M\_mc: 1024,\quad
  batch: 2048,\quad
  lambda\_hopf: 1e-2
\}
```

4 Entropy and Mixed Information Flow

We fix the EF neighborhood and prove a *coordinate-invariant* local equivalence between D_{KL} and the Fisher-metric norm of mixed cumulants. New in this draft: (i) an explicit radius via a Hessian-Lipschitz constant and sharp constants (Theorem 2); (ii) a conditioning corollary (Corollary 2); (iii) a windowed/transported whitening policy with bounds (Proposition 7).

4.1 Dictionary and EF model

Let $\mathcal{T}_K$ be a finite, centered, whitened dictionary of alternated mixed statistics up to total degree $K \in \{2, 3\}$ at *independence* $P_0 = P_\Gamma \otimes P_G$. Denote $F := \mathrm{FIM}(0) \succ 0$ with respect to $\mathcal{T}_K$. We use the EF

$$\frac{dP_\theta}{dP_0}(x) = \exp\left(\langle \theta, \mathcal{T}_K(x)\rangle - \psi(\theta)\right), \qquad \kappa_{\mathrm{mix}}(\theta) = \mathbb{E}_{P_\theta}[\mathcal{T}_K] = \nabla\psi(\theta).$$

Theorem 2 (FIM-invariant local equivalence)**.** [Inf] *Suppose ψ is twice differentiable on a neighborhood of 0 with $\nabla^2\psi(0) = F \succ 0$ and Hessian-Lipschitz with constant L_ψ there. For any $0 < \epsilon < 1$ and radius*

$$r \;\le\; \frac{\epsilon}{L_\psi},$$

we have, for all $\|\theta\| \le r$,

$$\frac{1}{2}(1-\epsilon)\,\|\kappa_{\mathrm{mix}}(\theta)\|^2_{F^{-1}} \;\le\; D_{\mathrm{KL}}(P_\theta\|P_0) \;\le\; \frac{1}{2}(1+\epsilon)\,\|\kappa_{\mathrm{mix}}(\theta)\|^2_{F^{-1}}.$$

Uses: (C5) $F \succ 0$, L_ψ, tolerance ϵ, radius $r \le \epsilon/L_\psi$.

Lemma 10 (Third-moment bound for L_ψ)**.** *For a whitened dictionary $\mathcal{T}_K$ and $F = \nabla^2\psi(0)$, there exist absolute $C > 0$ such that, for $\|u\| = 1$ and small $\|\theta\|$,*

$$\|\nabla^2\psi(\theta) - F\| \;\le\; C \sup_{\|u\|=1} \mathbb{E}_{P_0}\Big[\big|\langle u, T_K\rangle\big|^3\Big] \cdot \|\theta\| \;+\; O(\|\theta\|^2).$$

Hence one may take $L_\psi \lesssim \sup_{\|u\|=1} \mathbb{E}_{P_0}[|\langle u, T_K\rangle|^3]$.

Remark 3 (Estimator). Estimate the proxy via Hutch++/finite-difference probes (your `estimate_Lpsi` API) and log $(L_\psi,\ r = \epsilon/L_\psi)$ with a CI.

Remark 4 (Tight constants and how to pick them). The $1/2(1 \pm \epsilon)$ constants are optimal for quadratic ψ and remain sharp to first order in ϵ by Taylor's theorem with integral remainder and $\|\nabla^2\psi(\theta) - F\| \le L_\psi\|\theta\| \le \epsilon$.

Corollary 2 (Conditioning in unwhitened coordinates)**.** [Inf] *Let $\tilde\theta = R\theta$ for an invertible linear map with Fisher congruence $\tilde F = R^{-\top} F R^{-1}$. If $\chi = \kappa(F)$ is the condition number of F, then for the same ϵ and r,*

$$\frac{1}{2\chi}(1-\epsilon)\,\|\kappa_{\mathrm{mix}}(\tilde\theta)\|^2_{\tilde F^{-1}} \;\le\; D_{\mathrm{KL}}(P_{\tilde\theta}\|P_0) \;\le\; \frac{\chi}{2}(1+\epsilon)\,\|\kappa_{\mathrm{mix}}(\tilde\theta)\|^2_{\tilde F^{-1}}.$$

Uses: (C5) $\chi = \kappa(F)$; Fisher congruence under coordinate changes.

4.2 Windowed FIM and whitening across time

Let $\mu_{[t-\tau,t]}$ be the windowed law from Section 2. Denote by $F_t := \mathrm{FIM}_t(0)$ the Fisher matrix of the *same* dictionary $\mathcal{T}_K$ computed at independence within that window. Two whitening policies are admissible; pick one and use it consistently.

Definition 6 (Whitening policies)**.** *[*Inf/Stoch*]*

1. **Global whitening (baseline).** Fix $F := \mathrm{FIM}_{t_0}(0)$ at a reference window (or at the pooled baseline) and keep it fixed for all windows.
2. **Transported whitening (congruence).** For each window, whiten by F_t and transport coordinates to the baseline via the congruence map R_t defined by $F_t = R_t^{-\top} F R_t^{-1}$ (e.g., $R_t = F^{1/2} U_t \Lambda_t^{-1/2} U_t^\top$ with $F_t = U_t \Lambda_t U_t^\top$).

Define the *whitening leakage* $\chi(t) := \kappa(F_t F^{-1})$ and $\chi_{\max} := \sup_t \chi(t)$ on the time span of interest. *Budget & alarms:* enforce $\chi(t) \le \chi_{\max}$ (configuration) and log any exceedance; report $\chi_{\max}$ next to CI bars for time-series plots.

Uses: (C4) window length τ; (C5) baseline F, windowwise F_t, $\chi(t)$.

Proposition 7 (Window-local equivalence and transported bounds)**.** [Inf/Stoch] *Assume each window admits an EF neighborhood with the* same *Hessian-Lipschitz constant L_ψ for ψ_t on a fixed ball and that $F_t \succ 0$. Then, for either whitening policy in Definition 6:*

*1. (*Window-local equivalence*) For all $\|\theta\| \le r \le \epsilon/L_\psi$,*

$$\frac{1}{2}(1-\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F_t^{-1}} \;\le\; D_{\mathrm{KL}}(P^{(t)}_\theta \| P^{(t)}_0) \;\le\; \frac{1}{2}(1+\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F_t^{-1}}.$$

*2. (*Transport to baseline*) Expressing window-t cumulants in the baseline Fisher norm incurs at most $\chi(t)$ inflation:*

$$\frac{1}{2\,\chi(t)}(1-\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F^{-1}} \;\le\; D_{\mathrm{KL}}(P^{(t)}_\theta \| P^{(t)}_0) \;\le\; \frac{\chi(t)}{2}(1+\epsilon)\,\|\kappa^{(t)}_{\mathrm{mix}}(\theta)\|^2_{F^{-1}}.$$

If $\chi(t) \le \chi_{\max} < \infty$ uniformly (e.g. under slow drift and bounded transport dimension), then the constants are uniform in t with χ replaced by $\chi_{\max}$.

Uses: (C4) drift/mixing uniformity; (C5) F_t, L_ψ, $\chi(t)$.

Practical policy. Prefer *global whitening* when interpretability across time is crucial (fixed norm); prefer *transported whitening* when per-window conditioning is poor (reduces numerical ill-conditioning at the cost of tracking $\chi(t)$).

4.3 Diagnostics and parameter selection

Rule (Choosing ϵ and r). Pick $\epsilon \in [0.05, 0.2]$ to balance tightness vs. radius; set $r = \epsilon/L_\psi$ using a pilot bound on L_ψ from finite differences of $\nabla^2\psi$ at $\theta = 0$ (or Hutch++ trace bounds). Log (ϵ, r, L_ψ) alongside $\chi(t)$ for each window.

Acceptance checks for Section 4.

1. Every "local KL–cumulant" statement cites Theorem 2 with explicit (ϵ, r) and F.
2. Any computation outside whitened coordinates cites Corollary 2 (or Proposition 7 in windows) and reports the relevant χ.
3. Windowed analyses declare the whitening policy and either bound $\chi_{\max}$ or include it in reported uncertainty.

Numerical API (EF radius, L_ψ, whitening leakage $\chi(t)$)

```
# Assumptions
# - T(x): centered, whitened sufficient-statistic vector (same dictionary across windows)
# - Baseline Fisher F (SPD) is fixed (global whitening) or a chosen reference
# - window data: {x_i}_{i=1}^n with precomputed T_i := T(x_i)

LinearAlgebra:
  chol(SPD M) -> L s.t. M = L L^T
  solve_SPD(M, b) -> x with M x = b using Cholesky
  power_norm(A_mv, dim, iters=10) -> approx spectral norm via power iteration on mv-product
  hutchpp_opnorm(A_mv, dim, s=4, iters=2) -> Hutch++-style operator-norm proxy (optional)

# ---------- Window Fisher and whitening leakage ----------
function fisher_window(T_list):
    # Empirical Fisher at independence in this window (covariance of T)
    # ridge for numerical stability (lambda_ridge small, e.g. 1e-6 * trace/dim)
    n = len(T_list); d = dim(T_list[0])
    mu = mean(T_list)                          # should be ~0 by centering; keep for safety
    C  = (1/n) * sum_i (T_list[i] - mu) (T_list[i] - mu)^T
    lam = 1e-6 * trace(C) / d
    return C + lam * I_d

function chi_leakage(F_baseline, F_t):
    # Compute chi(t) = kappa(F_t F_baseline^{-1}) via generalized EVP
    # Equivalent to eigenvalues of S := F_baseline^{-1/2} F_t F_baseline^{-1/2}
    L = chol(F_baseline)                       # F = L L^T
    # mv for S: v -> L^{-1}^T * F_t * L^{-1} * v  (apply solves with L and L^T)
    function S_mv(v):
        y = solve_SPD(L, v)                    # L y = v
        z = F_t @ y
        w = solve_SPD(L.T, z)                  # L^T w = z
        return w
    # estimate extremal eigenvalues of S by power and inverse power
    # power_norm returns ||S||_2^{1/2} if fed S_mv? -> square it
    lam_max = power_norm(S_mv, dim=d, iters=12)^2
    # For lam_min, run power on inverse of S via CG solves (or Gershgorin fallback):
    # implement inverse-power via Rayleigh quotient iteration on (S + tau I)^{-1} if needed.
    # or use eigs on small d
    lam_min = 1.0 / power_norm(lambda v: cg_solve(S_mv, v), dim=d, iters=12)^2
    return lam_max / lam_min

# ---------- Small-tilt Hessian probes and L_psi estimation ----------
```

```
# EF identity: \nabla^2 psi(\theta) = Cov_{P_theta}(T). Around theta=0, estimate Lipschitz via:
#   || \nabla^2 psi(\delta u) - \nabla^2 psi(0) || / ||\delta u||.
# We approximate P_{delta u} by reweighting samples from P_0 with weights w_i \prop exp(\delta <u, T_i>

function weighted_cov(T_list, u, delta):
    s = [ dot(u, T_i) for T_i in T_list ]
    a = max(s)                          # log-sum-exp stabilization
    w_unn = [ exp(delta * (s_i - a)) for s_i in s ]
    Z  = sum(w_unn); w = [ w_unn_i / Z for w_unn_i in w_unn ]
    mu_w = sum_i w[i] * T_list[i]
    C_w  = sum_i w[i] * (T_list[i] - mu_w) (T_list[i] - mu_w)^T
    return C_w

function mv_diff_H(T_list, H0, u, delta):
    # returns an mv-product for A := (H(delta u) - H0)/delta without forming it
    H_delta = weighted_cov(T_list, u, delta)
    A = (H_delta - H0) / delta
    return (lambda v: A @ v)

function estimate_Lpsi(T_list, deltas=[1e-3, 3e-3, 1e-2], m_dirs=8, use_hutchpp=True):
    d  = dim(T_list[0])
    H0 = fisher_window(T_list)                   # \nabla^2 psi(0) at independence
    Lpsi_hat = 0.0
    for k in 1..m_dirs:
        u0 = gaussian(d); u = u0 / norm(u0)      # random unit direction (Euclidean or F-based)
        for delta in deltas:
            A_mv = mv_diff_H(T_list, H0, u, delta)  # mv for (H(delta u) - H0)/delta
            if use_hutchpp:
                opn = hutchpp_opnorm(A_mv, dim=d, s=4, iters=2)
            else:
                opn = power_norm(A_mv, dim=d, iters=10)
            Lpsi_hat = max(Lpsi_hat, opn)
    return Lpsi_hat, H0

# ---------- Radius selection and logging ----------
function pick_radius(epsilon, Lpsi_hat, floor=1e-8):
    Lp = max(Lpsi_hat, floor)
    r  = epsilon / Lp
    return r

function section5_online_update(window_T, F_baseline, epsilon, log_json):
    # 1) window Fisher and leakage
    F_t  = fisher_window(window_T)
    chi  = chi_leakage(F_baseline, F_t)
    # 2) Lipschitz estimate and EF radius
    (Lpsi_hat, H0) = estimate_Lpsi(window_T)
    r = pick_radius(epsilon, Lpsi_hat)
    # 3) structured logging (aligns with section 5 acceptance checks)
    log_json({
        "section": "5",
        "epsilon": epsilon,
        "Lpsi_hat": Lpsi_hat,
        "r_radius": r,
        "chi_t": chi,
```

```
        "trace_F_t": trace(F_t),
        "cond_F_t": cond(F_t),              # optional: standard cond estimate
        "n_stats": len(window_T),
        "d_dict": dim(window_T[0])
    })
    return {"F_t": F_t, "chi": chi, "Lpsi_hat": Lpsi_hat, "r": r, "H0": H0}

# ---------- Guards and unit checks ----------
# - If chi explodes (e.g., > 1e6), switch to transported whitening (section 5.2) and re-estimate.
# - If r is too small (below step size of optimizer), cap epsilon upward or enrich dictionary to reduce
# - Validate: for synthetic quadratic psi, Lpsi_hat -> 0 and r -> infty (subject to floor).
```

Notes and good practice.

- *Direction metric.* Draw u as unit vectors in the *Fisher* norm if you want L_ψ in the induced operator norm of F^{-1}; otherwise Euclidean is fine—just be consistent with the norm used in Theorem 2.
- *Small-tilt accuracy.* The reweighting step is accurate for small δ; keep $\delta \in [10^{-3}, 10^{-2}]$ (rescale if $\|T\|$ is large) and monitor ESS $= 1/\sum w_i^2$.
- *Stability.* Always ridge F_t lightly before Cholesky; report the ridge size in logs if it exceeds $10^{-6} \times \mathrm{trace}(F_t)/d$.
- *Throughput.* Cache T_i once per window; reuse them for both F_t and L_ψ estimation. Hutch++ gives a robust operator-norm proxy with a handful of matvecs.

References

[1] Jane Doe and John Smith. “An Example Reference for JSG”. In: *Journal of Examples* 42.1 (2020), pp. 1–10.